Fiscal Rules When Sovereign Debt Is Risky but Convenient

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Motivation: record debt, governed by simple rules

- Global public debt: over $100 trillion in 2024, and rising. View debt trends
- More than 100 countries have numerical fiscal rules (IMF).
  - EU: one **Maastricht-style** rule for all members, debt 60% and deficit 3% of GDP
  - Korea is considering a similar 60/3 rule
- **Most rules cap debt or deficits directly**, with the same cap for every member.
- **One rule, heterogeneous economies:** members differ in the alternative safe assets available to their investors. Where alternatives are scarce, government debt is the main safe asset; where they are abundant, it is not.

Sovereign debt has two faces

1. It is risky

- More debt ⇒ higher default risk and higher spreads (Eaton–Gersovitz 1981, Arellano 2008)
- Italy, 2001–2012: CDS premium of 79 bp on average
- The standard case *for* fiscal rules

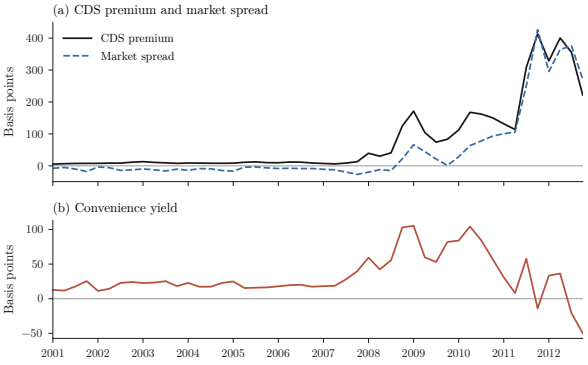
2. It is convenient

- Investors value government bonds for liquidity and as collateral, and accept a lower yield to hold them (Holmström–Tirole 1998, Krishnamurthy–Vissing-Jorgensen 2012)
- Italy, 2001–2012: convenience yield of 32 bp on average (Kaldorf–Röttger 2025)

An alternative: cap the spread

Spread brake (Hatchondo–Martinez–Roch 2022): stop borrowing when the spread passes a threshold, so the ceiling tracks default risk. **But which spread?**

$$\text{market spread} = \text{credit spread} - \text{convenience yield}$$



- **Market spread**: the yield over the risk-free rate that markets quote.
- **Credit spread**: compensation for default risk alone; in the data, the CDS premium.
- Italy's convenience yield: about 20 bp in 2001–07, 100 bp in 2008–10, negative in 2011–12.
- A market-spread brake would have responded to these movements too.

The convenience yield moves over time and across countries

- **Over time:** convenience yields move together across issuers, consistent with common demand for safe assets.
  - Italy, Spain, Portugal, Greece, 2001–07: common time effects explain about three quarters of the variance in convenience yields, controlling for each country's CDS premium.
  - The convenience yields rose together again in the 2008–10 flight to safety.
- **Across countries:** the convenience yield depends on the other safe collateral available to a country's investors.
  - Euro-area investors can use Bunds and other eligible collateral.
  - Where alternatives are scarce, the sovereign's own debt *is* the safe asset and its convenience yield is large.
- Either way, the market spread moves for reasons unrelated to the sovereign's default risk.

Research questions

Should a fiscal rule cap debt or cap spreads? If spreads, which spread? Can one rule serve heterogeneous economies?

This paper

What I do

- **Analytical model:** identifies the mechanisms: how the welfare effect of a fiscal rule decomposes, and how the market spread moves with demand for safe assets.
- **Quantitative analysis:** extends it to a calibrated infinite-horizon model and compares the debt brake, the credit-spread brake and the market-spread brake for Italy and for a union whose members differ only in access to outside safe collateral.

Preview of results

1. **Spread brakes** beat the best debt brake: their ceilings tighten when default risk rises. The cost: they are procyclical in recessions.
2. A brake on the **credit spread** beats one on the **market spread**, whose ceiling also moves with demand for safe assets.
3. In a **monetary union**, one credit-spread threshold is best for every member; a common debt ceiling or market-spread threshold is not.

Contribution

- Hatchondo, Martinez and Roch (2022) compare debt and spread brakes in a model whose spread reflects default risk only.

- **This paper** asks what changes when the spread also reflects demand for safe assets. Then a brake on the **credit spread** performs better: the market spread also moves with the convenience yield, which also moves for reasons other than default risk.

- **Policy implication for a monetary union:** unlike debt brakes and market-spread brakes, a credit-spread brake is portable across members, because the credit spread does not depend on each member's access to alternative safe collateral.

Related literature

- **Sovereign default and debt dilution:** Eaton–Gersovitz (1981), Arellano (2008), Chatterjee–Eyigungor (2012), Hatchondo–Martinez–Sosa-Padilla (2016), D'Erasmo–Mendoza (2016), Aguiar–Amador (2020, 2021).
- **Fiscal rules and commitment:** Amador–Werning–Angeletos (2006), Halac–Yared (2014, 2018), Liu–Shen (2022), Hatchondo–Martinez–Roch (2022), Alfaro–Kanczuk (2026).
- **Convenience and collateral value of government debt:** Holmström–Tirole (1998), Krishnamurthy–Vissing-Jorgensen (2012), van Binsbergen–Diamond–Grotteria (2022), Angeletos–Collard–Dellas (2023), Kaldorf–Röttger (2025).

Analytical model: environment

- The government issues one-period bonds B at $t = 0$ and repays or defaults at $t = 1$. Revenue is stochastic; default is a low-income event.
- **Domestic holders** own a share s_h of the debt: repayment is partly a transfer to domestic residents.
- **Collateral services:** domestic holders value the services of the bonds they hold, so convenience enters welfare and not only the bond price.
- **Political myopia:** the government discounts at β_G, society at $\beta_S \geq \beta_G$, so the government borrows more than society would choose.

$$\mathcal{W}(B;\beta) = \underbrace{\psi_g v(g_0) - s_h q(B) B}_{\text{today}} + \beta \underbrace{s_h \zeta\, u\big(\Theta(B)\big)}_{\text{collateral services}} + \beta \underbrace{\mathbb{E}_y \max\left\{V^R(B,y), V^D(y)\right\}}_{\text{repay or default}} \tag{3}$$

The government chooses $B^G = \arg\max_B \mathcal{W}(B;\beta_G)$; society evaluates $\mathcal{W}_S(B) = \mathcal{W}(B;\beta_S)$.

Details

Analytical model: pricing risky but convenient debt

$$q(B) = \frac{\overbrace{\big[1 - p(B)\big]}^{\text{credit value}} \quad \overbrace{\big[1 + \Lambda(B)\big]}^{\text{convenience markup}}}{1 + r_f} \tag{5}$$

- The convenience yield Λ is the pledgeable fraction of the bond times the marginal value of collateral, $\Lambda = \big[1 - \kappa(p)\big]\, \zeta\, u'(\Theta)$.
- **Risk lowers the bond's value twice:** as a credit claim $(1 - p)$ and as collateral, through a haircut $\kappa(p)$ that rises with default risk.
- **Demand for safe assets** ζ raises the convenience yield at every level of debt.
- **Outside safe collateral** b_{alt} enters total collateral Θ: more of it makes collateral less scarce and lowers Λ.
- **Credit spread:** $p(B)$, compensation for default risk alone. **Market spread:** the yield over the risk-free rate, about $p(B) - \Lambda(B, \zeta)$: default risk net of the convenience yield.

▸ Default ▸ Pricing ▸ Collateral

Proposition 1: welfare decomposition

When a debt ceiling binds, the welfare effect of loosening it could be decomposed into three channels:

$$\boxed{\frac{d\mathcal{W}_S}{d\bar{B}} = S + D + L} \tag{9}$$

$D < 0$ when more debt raises default risk. Without convenience, $L = 0$.

- S**, smoothing:** more public goods today against more repayment tomorrow, net of the transfers to domestic holders.
- D**, default:** the marginal bond raises default risk and lowers the price of the whole stock. In a standard default model this favors a tighter rule.
- L**, liquidity:** the convenience revenue on the marginal bond, the change in the convenience yield on the stock, and the collateral services to domestic holders.

▸ Formulas

Proposition 2: political myopia and the value of a debt brake

At the government's choice B^G:

$$\left.\frac{d\mathcal{W}_S}{dB}\right|_{B^G} = (\beta_S - \beta_G)\Big[\underbrace{s_h\,\zeta\,u'(\Theta)\,\theta'(B^G)}_{\text{collateral services}} - \underbrace{\mathbb{E}\big[(1-d)\{\psi_g v'(g_1) - s_h\}\big]}_{\text{repayment burden}}\Big] \tag{13}$$

- $\beta_S = \beta_G$: B^G is socially optimal — no rule can raise welfare.
- $\beta_S > \beta_G$: if the repayment burden exceeds collateral services, a cap $\bar{B} < B^G$ raises welfare. The optimal cap solves $S + D + L = 0$.

Where outside safe collateral enters

Scarce outside collateral (low b_{alt}) raises $u'(\Theta)$. If $\theta'(B) > 0$, this enlarges the collateral term and **shrinks the gain from a cap**: society is reluctant to cut debt that serves as scarce collateral.

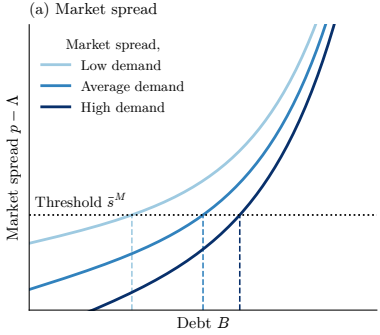
▸ Derivation

Proposition 3: market-spread vs. credit-spread brake

A market-spread brake with threshold $\bar{s}^M$ permits debt B if and only if

$$\underbrace{p(B) - \Lambda(B,\zeta)}_{\text{market spread}} \leq \bar{s}^M \iff \boxed{\underbrace{p(B)}_{\text{credit spread}} \leq \bar{s}^M + \Lambda(B,\zeta)} \tag{14}$$

- When demand for safe assets is high, Λ is high, so the brake lets the government borrow more, even though default risk is the same.



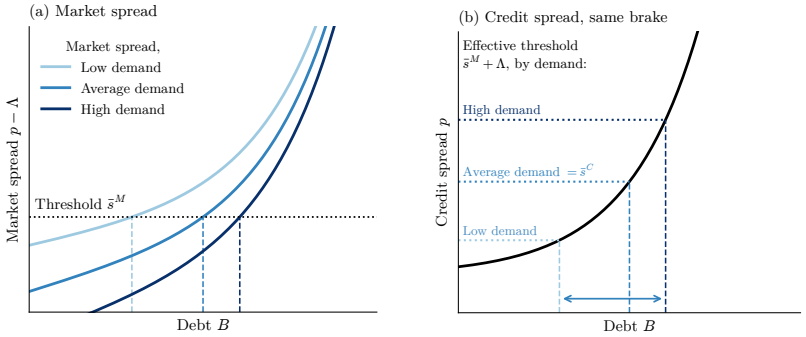
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Details

From the analytical model to the quantitative model

The analytical model signs the channels and shows one reason for a rule: **political myopia** ($\beta_G < \beta_S$). The quantitative model adds a second friction and puts numbers on both.

1. **Infinite horizon, Markov discretion.** The government re-optimizes every quarter and cannot commit to its future issuance.
2. **Long-term debt** ($\delta = 0.05$, five-year maturity). Today's bond price embeds its future *resale* value $\Rightarrow$ future issuance **dilutes** legacy bondholders (Aguiar–Amador 2021).
3. **Stochastic environment calibrated to Italy:** income and demand for safe assets ζ follow AR(1)s; default entails an output loss, exclusion and restructuring.

Two frictions a rule corrects

Myopia (analytical and quantitative) and **dilution** (quantitative only). Preserved: the pricing $q = m(1+\Lambda)/(1+r_f)$ and the channels (S), (D), (L).

Fiscal rules and welfare

Rules follow Hatchondo–Martinez–Roch (2022): a ceiling on next-quarter face debt; a government above it may keep its debt but not increase it.

Debt brake — a fixed level cap

$$B' \leq \max\{\bar{B},\ B\}$$

Spread brake — a cap that moves with risk

$$B' \leq \max\{\bar{B}_s(y),\ B\}$$

$$\bar{B}_s(y) = \max\{B' :\ \text{spread}(B', y) \leq \bar{s}\} \qquad (32)$$

- **Market-spread brake:** the yield over the risk-free rate on the new bond: default risk net of the convenience yield.
- **Credit-spread brake:** the same, for the bond priced without convenience value: default risk alone, the model counterpart of the CDS premium.
- **Welfare:** consumption equivalent, the permanent % increase in private consumption that makes society as well off without the rule as with it.

▸ Pricing

Calibration: Italy, 2001–2012

Parameter	Description	Value	Source or target
Set outside the model			
$\rho_y,\ \sigma_y$	Income persistence, volatility	0.937, 0.0092	Italian GDP
$\delta,\ \chi$	Maturity, coupon	0.05, 0.0115	Five-year maturity; average coupon
$\omega,\ \vartheta$	Recovery, re-entry	0.63, 0.08	Cruces–Trebesch (2013)
$\mu,\ \bar{\kappa}$	Haircut elasticity, maximum haircut	0.4, 0.7	Eurosystem collateral framework
s_h	Domestic share of debt	0.505	Private domestic holdings
b_{alt}	Outside safe collateral	4.34	German debt / Italian quarterly GDP
ζ_2	Scarcity slope	0.20	Panel of euro-area sovereigns
β_S	Social discount factor	0.99	Hatchondo et al. (2022)
Calibrated jointly (9 parameters, 11 moments)			*Most informative moment*
β_G	Government discount factor	0.9807	Mean debt
$\psi_g,\ \bar{g}$	Public goods: weight, subsistence	1.9×10^{-4}, 0.17	Spending / GDP and its cyclicality
$\lambda_0,\ \lambda_1$	Output loss in default	-0.64, 0.74	CDS mean and volatility, output loss
$\bar{\zeta},\ \sigma_\zeta,\ \rho_\zeta$	Demand for safe assets	0.0045, 0.74, 0.84	Convenience yield: mean, calm sd, persistence
σ_ε	Scale of taste shocks	0.013	Volatility of debt

The volatility of demand is calibrated to the calm years 2001–2007, when convenience yields moved together across countries and default risk was low. The model's credit spread, the spread on the same bond without convenience value, is matched to the CDS premium.

▸ All parameters

Calibration: model fit

Moment	Data	Model
Targeted		
Mean debt / quarterly GDP	4.59	4.62
Mean CDS premium (bp)	79	83
Volatility of CDS premium (bp)	113	102
Mean convenience yield (bp)	32	31
Volatility of convenience yield, calm (bp)	5.8	5.7
Public spending / GDP	0.193	0.188
Output loss in default	0.10	0.08
Not targeted		
Mean market spread (bp)	47	52
Volatility of convenience yield, full sample (bp)	31	6
Default frequency (% per year)	–	1.82

The full-sample volatility of the convenience yield comes almost entirely from 2008–2012: the 2008–10 flight to safety and Italy's own fall in 2011–12. These episodes are studied separately. ▸ Episodes

Result 1: spread brakes beat the best debt brake

	No rule	Debt brake (101.2%)	Credit-spread brake (7.5 bp)	Market-spread brake (−12.5 bp)
Welfare gain (% of consumption)	–	**0.58**	**0.85**	**0.82**
Mean debt (% of GDP)	101.1	96.9	116.5	116.8
Default (% per year)	1.82	0.03	0.17	0.26
Mean credit spread (bp)	83.1	1.3	6.6	10.5
Mean convenience yield (bp)	31.1	30.4	25.5	25.4
Share of states where the rule binds	–	0.62	0.53	0.51

- Every rule lowers default risk. The debt brake does it by holding debt down in good times as well as bad.
- The credit-spread brake's ceiling moves with income: it binds when default risk rises, so debt can rise to 117% of GDP at a credit spread of 7 bp.

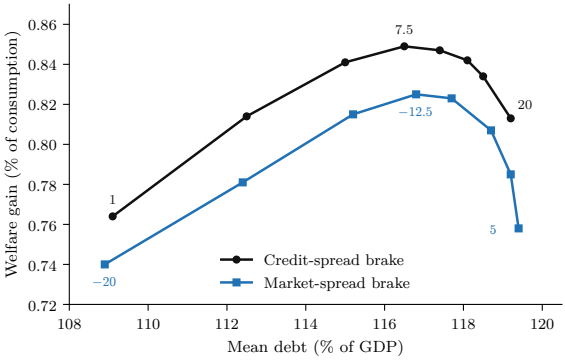
Result 1: where the gains come from

	Debt	Credit	Market
Welfare gain	0.58	0.85	0.82
S: smoothing	−0.58	−0.33	−0.30
D: default	1.01	1.04	0.99
price of debt	0.55	0.59	0.57
output loss	0.47	0.44	0.42
L: liquidity	0.02	0.06	0.06
Residual	0.13	0.08	0.08

- **Lower default risk drives every gain.**
- The price gain in D and the loss in S largely offset: lenders are risk neutral, so a higher price is paid for by repaying in states that would have defaulted.
- Debt brake: 0.55 against −0.58, so its gain is the avoided output loss.
- Credit-spread brake: a further 0.26 from D and S together, because it borrows more when risk is low.
- L **is small under every rule**, but the convenience yield decides which spread the brake should read.

▸ How the channels are measured

Result 2: the credit-spread brake beats the market-spread brake



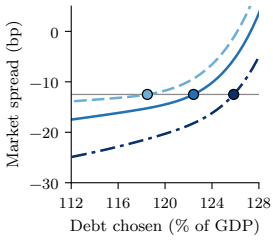
- Thresholds on different spreads cannot be compared directly, so each threshold is plotted against the mean debt it produces.
- The credit-spread brake does better at **every level of mean debt**.
- At the best thresholds: 0.85 against 0.82% of consumption.

Result 2: why — Proposition 3 in the quantitative model

Proposition 3: $\underbrace{p(B) - \Lambda(B,\zeta)}_{\text{market spread}} \leq \bar{s}^M \iff \underbrace{p(B)}_{\text{credit spread}} \leq \bar{s}^M + \Lambda(B,\zeta)$ (14)

Low demand — Median demand — High demand

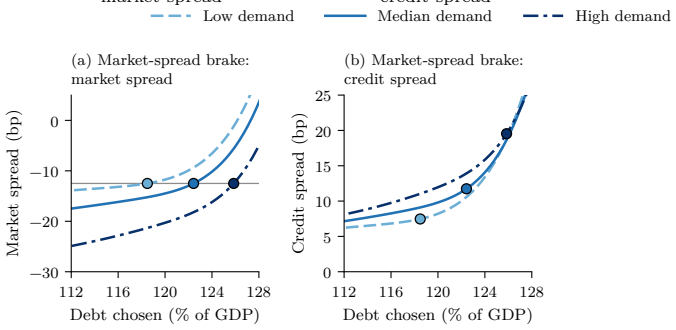
(a) Market-spread brake: market spread



- **(a) Market-spread brake** (−12.5 bp): higher demand, lower market spread; ceiling 118.5 to 125.9% of GDP.

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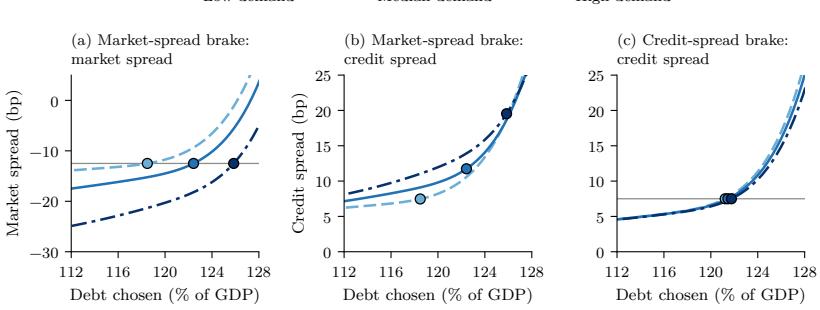
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- **(a) Market-spread brake** (−12.5 bp): higher demand, lower market spread; ceiling 118.5 to 125.9% of GDP.
- **(b) Same brake, in credit spreads:** threshold $\bar{s}^M + \Lambda$ rises from 7.5 to 19.5 bp.

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- **(b) Same brake, in credit spreads:** threshold $\bar{s}^M + \Lambda$ rises from 7.5 to 19.5 bp.
- **(c) Credit-spread brake** (7.5 bp): fixed threshold; ceiling 121.2 to 121.8%.

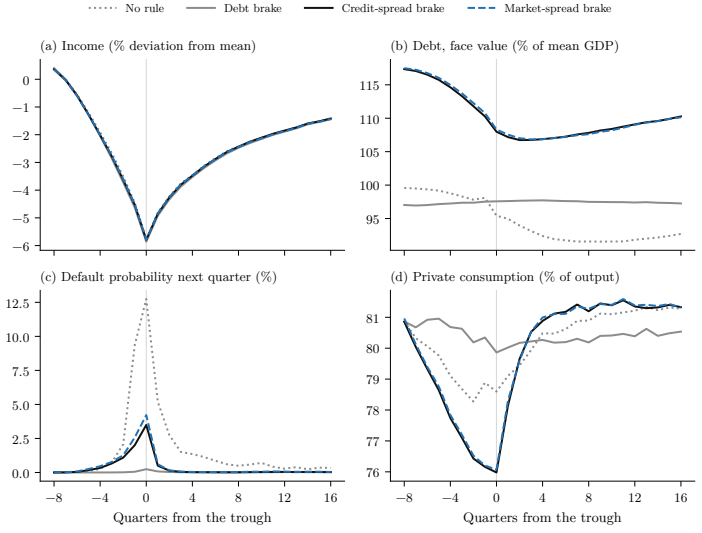
Result 2: loose when demand is high, tight when it is low

	Credit-spread brake			Market-spread brake		
Demand for safe assets	Low	Median	High	Low	Median	High
Effective credit threshold (bp)	7.5	7.5	7.5	7.6	12.2	20.5
Ceiling (% of GDP)	121.6	121.8	122.1	118.6	122.6	126.1
Credit spread on debt (bp)	6.7	6.6	6.5	7.8	10.3	13.5
Share of states above the ceiling	0.19	0.18	0.17	0.26	0.17	0.12

- High demand: the market-spread brake lets the government borrow at higher default risk.
- Low demand: it forces the government to stop borrowing although default risk has not risen.
- Over the cycle, the market-spread brake gains in median- and high-demand quarters (+0.03, +0.11% of consumption) but loses more in low-demand quarters (−0.17), when it must repay the extra debt: −0.03 in total.
- A larger fall in demand moves its ceiling much further: by 38 points of GDP on impact.

▸ Episodes

Result 3: the cost of a spread brake — recessions



- **Debt:** the spread brakes let debt rise in normal times: 117% of GDP before a recession, against 100 without a rule and 97 under the debt brake.
- **Ceilings:** as income falls, default risk rises and their ceilings fall.
- **Private consumption** bears the adjustment: 81 to 76% of output at the trough, against 79 without a rule and 80 under the debt brake.
- **Default:** 12% of these recessions end in default under the credit-spread brake, against 39% without a rule.

Result 3: gains by income

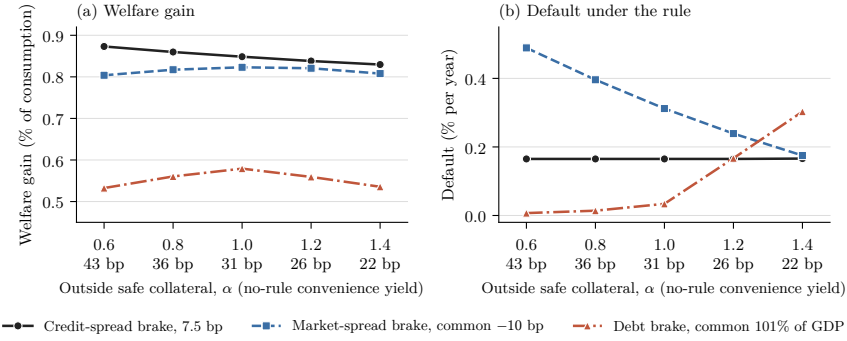
Income (sd from mean)	< -1.5	-1.5 to -0.5	-0.5 to 0.5	0.5 to 1.5	> 1.5	Total
Share of quarters	0.07	0.25	0.35	0.25	0.07	1
Debt brake	0.15	0.36	0.15	-0.05	-0.03	0.58
Credit-spread brake	0.02	0.16	0.30	0.27	0.09	0.84
Market-spread brake	0.01	0.17	0.30	0.25	0.08	0.81
Credit − debt brake	-0.13	-0.19	0.15	0.32	0.12	0.26

- The debt brake gains most in low-income quarters, when default would otherwise occur.
- The credit-spread brake gains most in average and good times, when its ceiling lets the government borrow at low spreads.
- Against the debt brake: -0.33 in the lowest-income third of quarters, $+0.58$ in the rest.

Result 4: one rule for a monetary union

Members identical to Italy except outside collateral, $b_{\text{alt}} = 4.34\,\alpha$, $\alpha \in [0.6, 1.4]$.

▸ Table



- **Credit-spread brake:** 7.5 bp is best for every member; all default 0.17% per year.
- **Common market-spread brake:** more default where the convenience yield is higher (0.49% vs. 0.17%); costs up to 0.07%.
- **Common debt ceiling:** too tight where collateral is scarce, too loose where abundant.

Conclusion

1. **Cap the spread, not the level of debt.** A brake on the credit spread is worth 0.85% of consumption against 0.58% for the best debt brake. Its ceiling moves with default risk.
2. **The flexibility has a cost.** The credit-spread brake enters recessions with more debt, and private consumption falls more when its ceiling tightens.
3. **Apply the threshold to default risk.** The market spread also moves with demand for safe assets: 0.02% lost at the benchmark, up to 0.07% in a union, where one market-spread threshold is a different default-risk threshold for each member.

Appendix: robustness to society's discount factor

	$\beta_S = 0.985$	$\beta_S = 0.99$	$\beta_S = 0.995$
Credit-spread brake minus best debt brake	0.51	0.27	0.01
Credit-spread minus market-spread brake	0.02–0.03	0.02–0.03	0.02–0.03

- The credit-spread brake is the best rule at every β_S.
- A more patient society gains more from holding debt down, which favors the debt brake. At $\beta_S = 0.995$ the rules are close: 0.65 (credit), 0.64 (debt), 0.62 (market).
- The ranking of the two spread brakes does not depend on β_S.

Appendix: limitations

- Demand volatility is calibrated to the calm years; the model does not produce a crisis like 2011–12, when demand for Italian debt and default risk moved together.
- Union members take outside collateral as given, although it includes other members' bonds.

Appendix: environment and objectives

$$g_0 = \tau y_0 + q(B)B, \qquad g_1^R = \tau y_1 - B, \qquad g_1^D = \tau y_1, \tag{1}$$

$$V^R(B, y) = \psi_g v(\tau y - B) + s_h B, \qquad V^D(y) = \psi_g \big[v(\tau y) - \phi(\tau y)\big]. \tag{2}$$

- **Domestic holders.** $-s_h q(B)B$ is the domestic cost of financing current issuance; $+s_h B$ in V^R means repayment is partly a transfer back to residents.
- **Collateral services.** $\beta\, s_h\, \zeta\, u(\Theta)$ is the value of the services obtained while the bond is held. It is common to both branches because it accrues during the holding period.
- **Default** wipes out the debt and costs $\psi_g \phi(\tau y)$ in utility.
- **Myopia.** Repay or default is decided at $t = 1$, so p, q and Θ do not depend on β_G or β_S; only the valuation of the future does.

Back

Appendix: default incentives with domestic holders

Define $\Delta(B, y) \equiv V^D(y) - V^R(B, y)$. Since v is concave and $\phi' \geq 0$,

$$\Delta_y = \psi_g \tau \Big[v'(\tau y) - v'(\tau y - B) - \phi'(\tau y) \Big] < 0,$$

so default is a **low-income event**: $d = 1$ iff $y < y^*(B)$, and $p(B) = F(y^*(B))$.

$$\frac{dy^*}{dB} = \frac{\psi_g v'(\tau y^* - B) - s_h}{|\Delta_y(B, y^*)|}, \qquad p'(B) = f(y^*) \frac{dy^*}{dB}. \tag{4}$$

Interpretation

More debt raises default incentives when $\psi_g v'(\tau y^* - B) > s_h$. Domestic ownership weakens the link between debt and default: part of repayment is an internal transfer.

Back

Appendix: pricing risky but convenient debt

Investors are risk neutral and value the bond as collateral. With zero recovery,

$$q(B) = \frac{[1 - p(B)]\,[1 + \Lambda(B)]}{1 + r_f}, \tag{5}$$

$$\begin{aligned} \theta(B) &= \big[1 - \kappa(p)\big](1 - p)B && \text{usable sovereign collateral} \\ \Theta(B) &= \theta(B) + b_{\text{alt}} && \text{total collateral} \\ \Lambda(B) &= \big[1 - \kappa(p)\big]\,\zeta\,u'\big(\Theta(B)\big) && \text{convenience yield} \end{aligned} \tag{6}$$

- Default risk enters twice: through $1 - p$ and through the haircut $\kappa(p)$.
- Outside collateral b_{alt} affects the price only through $u'(\Theta)$.
- **Spreads:** the yield over the risk-free rate is $\frac{1/q}{1+r_f} - 1 = \frac{1}{(1-p)(1+\Lambda)} - 1 \approx p - \Lambda$, the market spread. Without convenience value ($\Lambda = 0$) it is $\approx p$, the credit spread.

Kaldorf–Röttger (2025)

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Appendix: debt, usable collateral and the convenience yield

$$\theta'(B) = \underbrace{\big[1-\kappa(p)\big](1-p)}_{\text{quantity}} - \underbrace{Bp'(B)\Big[\kappa'(p)(1-p)+1-\kappa(p)\Big]}_{\text{quality}} \tag{7}$$

If quality deteriorates enough, $\theta'(B) < 0$: more face debt, less usable collateral.

$$\Lambda'(B) = \underbrace{\big[1-\kappa(p)\big]\,\zeta\,u''(\Theta)\,\theta'(B)}_{\text{collateral scarcity}} - \underbrace{\kappa'(p)\,p'(B)\,\zeta\,u'(\Theta)}_{\text{collateral haircut}} \tag{8}$$

- **Scarcity:** if debt adds usable collateral, collateral becomes less scarce and Λ falls.
- **Haircut:** more debt raises default risk and the haircut, which lowers Λ.
- Through θ', default risk can also make collateral scarcer and raise Λ.

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Appendix: the three channels

$$S = \big[\psi_g v'(g_0) - s_h\big]\frac{1-p}{1+r_f} - \beta_S\,\mathbb{E}\big[(1-d)\{\psi_g v'(g_1) - s_h\}\big] \tag{10}$$

$$D = -\big[\psi_g v'(g_0) - s_h\big]\frac{\bar{B}\,p'(\bar{B})(1+\Lambda)}{1+r_f} \tag{11}$$

$$L = \big[\psi_g v'(g_0) - s_h\big]\frac{(1-p)\big[\Lambda + \bar{B}\Lambda'(\bar{B})\big]}{1+r_f} + \beta_S\, s_h\, \zeta\, u'(\Theta)\,\theta'(\bar{B}) \tag{12}$$

Derivation.

1. Today's terms change by $[\psi_g v'(g_0) - s_h]\, d(qB)/dB$.
2. By the envelope argument, the continuation value changes by $-\mathbb{E}\big[(1-d)\{\psi_g v'(g_1) - s_h\}\big]$.
3. Marginal revenue: $d(qB)/dB = \big[(1-p)(1+\Lambda) - Bp'(1+\Lambda) + (1-p)B\Lambda'\big]/(1+r_f)$.
4. $(1-p)$ joins S; $(1-p)\Lambda$ and $(1-p)B\Lambda'$ join L; $-Bp'(1+\Lambda)$ is D; add the collateral services $\beta_S s_h \zeta u'(\Theta)\theta'$ to L.

The government's condition is the same with β_G in place of β_S.

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Appendix: derivation of Proposition 2

Society and the government evaluate the same objective (3), which is linear in the discount factor:

$$\mathcal{W}(B;\beta) = \underbrace{\psi_g v(g_0) - s_h q(B) B}_{T(B)} + \beta \underbrace{\Big[s_h\, \zeta\, u\big(\Theta(B)\big) + \mathbb{E}_y \max\{V^R, V^D\} \Big]}_{F(B)}. \tag{3}$$

Since $\partial \mathcal{W}(B^G;\beta_G)/\partial B = 0$ at the government's choice,

$$\left.\frac{d\mathcal{W}_S}{dB}\right|_{B^G} = \frac{d}{dB}\Big[\mathcal{W}(B;\beta_S) - \mathcal{W}(B;\beta_G)\Big]_{B^G} = (\beta_S - \beta_G)\, F'(B^G).$$

- Today's terms T cancel; only the future terms remain, which society weighs more.
- $F'(B) = s_h \zeta u'(\Theta)\theta'(B) - \mathbb{E}\big[(1-d)\{\psi_g v'(g_1) - s_h\}\big]$, by the envelope theorem at the default cutoff. This gives (13).
- **Outside collateral.** Holding debt fixed, $\partial\big[s_h \zeta u'(\Theta)\theta'\big]/\partial b_{\text{alt}} = s_h \zeta u''(\Theta)\theta'$, negative when $\theta' > 0$; the sign reverses if $\theta' < 0$. This is a partial effect; the quantitative model computes the total effect.

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Appendix: how far the market-spread ceiling moves

A market-spread brake stops borrowing where the credit spread reaches the effective threshold

$$\bar{s}^M + \Lambda\big(\bar{B}^M, \zeta\big). \tag{14}$$

Its ceiling moves with demand by

$$\frac{d\bar{B}^M}{d\zeta} = \frac{\partial\Lambda/\partial\zeta}{p'(\bar{B}^M) - \partial\Lambda/\partial B}, \tag{15}$$

the shift in the market-spread curve divided by its slope.

- It is equivalent to a credit-spread brake with a constant threshold $\bar{s}^C$ if and only if the convenience yield at its ceiling, $\bar{\Lambda}$, is the same in every state; then $\bar{s}^C = \bar{s}^M + \bar{\Lambda}$.
- A credit-spread brake with a constant threshold has the same ceiling at every level of demand.
- The move helps only if the best ceiling moves with demand by about as much. Nothing ties the two.

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Appendix: pricing and spreads in the quantitative model

$$q(B', s) = \frac{m(B', s)\left[1 + \Lambda(B', s)\right]}{1 + r_f} \tag{26}$$

- m: expected payoff of a new bond, including its **resale value** next quarter. A government that will issue more tomorrow lowers today's price (dilution).
- Λ: the pledgeable fraction times the marginal value of collateral, which falls with total collateral at the rate ζ_2 (the scarcity slope).
- **Credit price** q_P: the price of a bond with the same default risk and no convenience value; it solves (23) and (26) with $\Lambda = 0$ and q_P in place of q.

For any price, the annualized spread over the risk-free rate, in basis points, is

$$\mathrm{spr}(q) = 10^4\left[\left(1 + \nu/q - \delta\right)^4 - (1 + r_f)^4\right]. \tag{28}$$

- ν: coupon; δ: share of debt maturing each quarter.
- Credit spread $= \mathrm{spr}(q_P)$, the model counterpart of the CDS premium. Market spread $= \mathrm{spr}(q)$. Convenience yield $= \mathrm{spr}(q_P) - \mathrm{spr}(q)$.

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Appendix: the quantitative model

Net borrowing $R = q(B', s)\big[B' - (1-\delta)B\big] - \nu B$; budgets $g = T + R$, $c = y - T - s_h R$.

$$V^R(B, B', s) = U(R, y) + \beta_G\, h\big(\Theta(B', s), \zeta\big) + \beta_G\, \mathbb{E}_{s'|s} V(B', s') \tag{29}$$

$$V^D(B, s) = U_d(y) + \beta_G\, h\big(\Theta_d, \zeta\big) + \beta_G\, \mathbb{E}_{s'|s}\big[\vartheta V(\omega B, s') + (1-\vartheta) V^D(B, s')\big] \tag{30}$$

$$V(B, s) = \max\Big\{V^D(B, s),\ \max_{B' \in \Gamma(B,s)} V^R(B, B', s)\Big\} \tag{31}$$

$$m(B', s) = \mathbb{E}_{s'|s}\Big[(1-d')\big\{\nu + (1-\delta)\, q(B'', s')\big\} + d'\, q_d(B', s')\Big] \tag{23}$$

$$\Theta(B', s) = \big[1 - \kappa(B', s)\big]\, m(B', s)\, B' + b_{\text{alt}}, \qquad \Lambda = \big[1-\kappa\big]\, \zeta\, e^{-\zeta_2 \Theta} \tag{24–25}$$

- Output loss in default $L(y) = \max\{0, \lambda_0 y + \lambda_1 y^2\}$; haircut $\kappa = \min\{p^\mu, \bar{\kappa}\}$.
- Markov equilibrium; small taste shocks on each option.

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Appendix: all parameters

	Description	Value	Source or target
Set outside the model			
$\rho_y,\ \sigma_y$	Income persistence, volatility	0.937, 0.0092	Italian GDP
r_f	Risk-free rate (quarterly)	0.0027	Real swap rate
$\delta,\ \chi$	Maturity, coupon	0.05, 0.0115	Five-year maturity; average coupon
ω	Recovery	0.63	Cruces–Trebesch (2013)
ϑ	Re-entry probability	0.08	Three-year exclusion
$\mu,\ \bar{\kappa}$	Haircut elasticity, maximum haircut	0.4, 0.7	Eurosystem collateral framework
s_h	Domestic share of debt	0.505	Private domestic holdings
b_{alt}	Outside safe collateral	4.34	German debt / Italian quarterly GDP
ζ_2	Scarcity slope	0.20	Panel of euro-area sovereigns
$\gamma,\ \sigma$	Curvature, public and private consumption	2, 2	Kaldorf–Röttger; Hatchondo et al.
θ_c	Weight on private consumption	0.639	$\theta_c u_c'(\bar{c}) = 1$
β_S	Social discount factor	0.99	Hatchondo et al. (2022)
Calibrated jointly			*Most informative moment*
β_G	Government discount factor	0.9807	Mean debt
ψ_g	Weight on public goods	1.89×10^{-4}	Public spending / GDP
$\bar{g}$	Subsistence public goods	0.1729	Cyclicality of public spending
λ_0	Output loss, linear term	-0.642	Output loss; CDS mean and volatility
λ_1	Output loss, quadratic term	0.737	Jointly with λ_0
$\bar{\zeta}$	Mean demand for safe assets	0.00446	Mean convenience yield
σ_ζ	Volatility of demand	0.743	Calm volatility of convenience yield
ρ_ζ	Persistence of demand	0.844	Persistence of German convenience yield
σ_ε	Scale of taste shocks	0.0131	Volatility of debt

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Appendix: measuring the channels in the quantitative model

- S corresponds to the change in net borrowing from foreign investors, valued at the credit prices of the economy without a rule.
- D corresponds to the higher price lenders pay because default risk is lower, plus the output losses avoided by defaulting less.
- L corresponds to the convenience revenue on the debt issued plus the collateral services of the debt held at home.
- Each change in resources is valued at the marginal utility of consumption in the quarter and state in which it occurs, then converted into consumption equivalents.
- **Residual:** the channels value mean resources in each quarter and state. A rule that makes consumption less dispersed within them raises welfare even when mean resources do not change.

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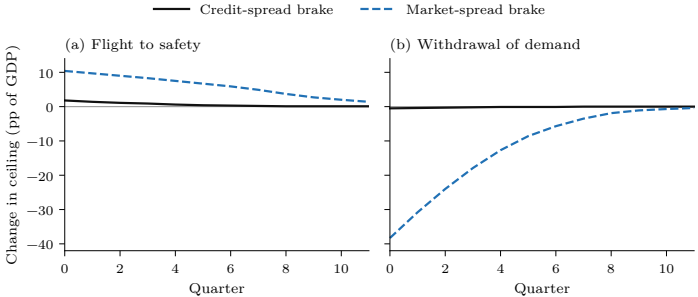
Appendix: the union in numbers

Outside collateral, α	0.6	0.8	1.0	1.2	1.4
No rule					
Convenience yield (bp)	42.8	36.5	31.1	26.4	22.4
Debt (% of GDP)	103.4	102.1	101.1	100.2	99.4
Welfare gain (% of consumption)					
Credit-spread brake, 7.5 bp	0.87	0.86	0.85	0.84	0.83
Market-spread brake, common (−10 bp)	0.80	0.82	0.82	0.82	0.81
Market-spread brake, own best	0.83	0.83	0.82	0.82	0.82
Debt brake, common (101.2%)	0.53	0.56	0.58	0.56	0.54
Debt brake, own best	0.58	0.59	0.58	0.57	0.57
Best threshold for each member					
Market-spread brake (bp)	−20	−15	−12.5	−10	−7.5
Debt brake (% of GDP)	103.5	102.7	101.2	99.7	99.7
Default under the rule (% per year)					
Credit-spread brake, 7.5 bp	0.17	0.17	0.17	0.17	0.17
Market-spread brake, common	0.49	0.40	0.31	0.24	0.17
Debt brake, common	0.01	0.01	0.03	0.17	0.30

Even a member-specific market-spread threshold falls short of the credit-spread brake: a constant threshold cannot track a convenience yield that moves with demand over time.

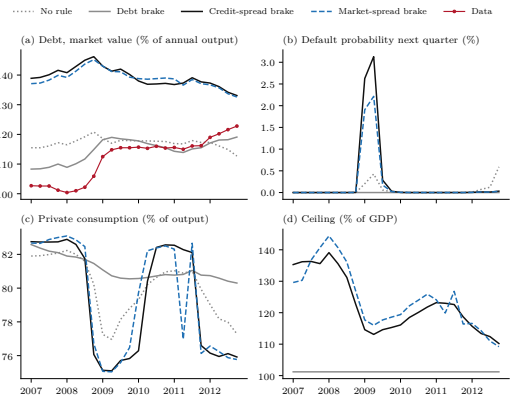
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Appendix: large temporary shifts in demand

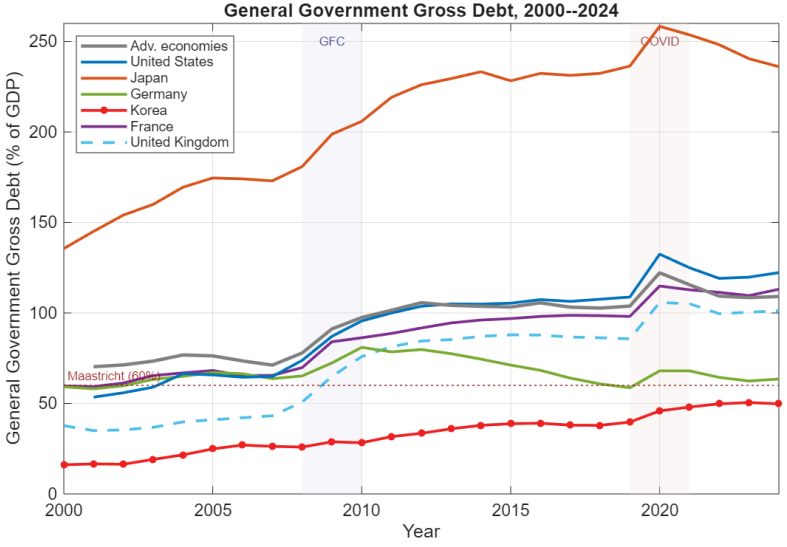


- Unanticipated paths of demand for eight quarters, known once they begin.
- **Flight to safety** (convenience yield +30 bp): the market-spread ceiling loosens by 8 points, but the government does not borrow against temporary cheap funding. The gap between the brakes barely changes (+0.001).
- **Withdrawal of demand** (demand to zero): the market-spread ceiling falls by 38 points on impact and the brake stops all new borrowing; the credit-spread ceiling does not move. The gap widens by 0.014%.

Appendix: Italy, 2007–2012



- Each economy under Italy's income and demand shocks of 2007Q1–2012Q4.
- Adopted in 2007Q1, the ranking holds: credit-spread brake 0.86, market-spread brake 0.83, debt brake 0.65% of consumption.
- In 2009 and 2011–12 the spread brakes bind and private consumption falls to 75–76% of output, against 77 to 80 without a rule and 80 to 81 under the debt brake.
- The model's spreads stay well below Italy's in 2011–12: it does not reproduce the sovereign-debt crisis.



Back to motivation